U O W

CSIT314
Software Development Methodologies

Data-driven Software Development

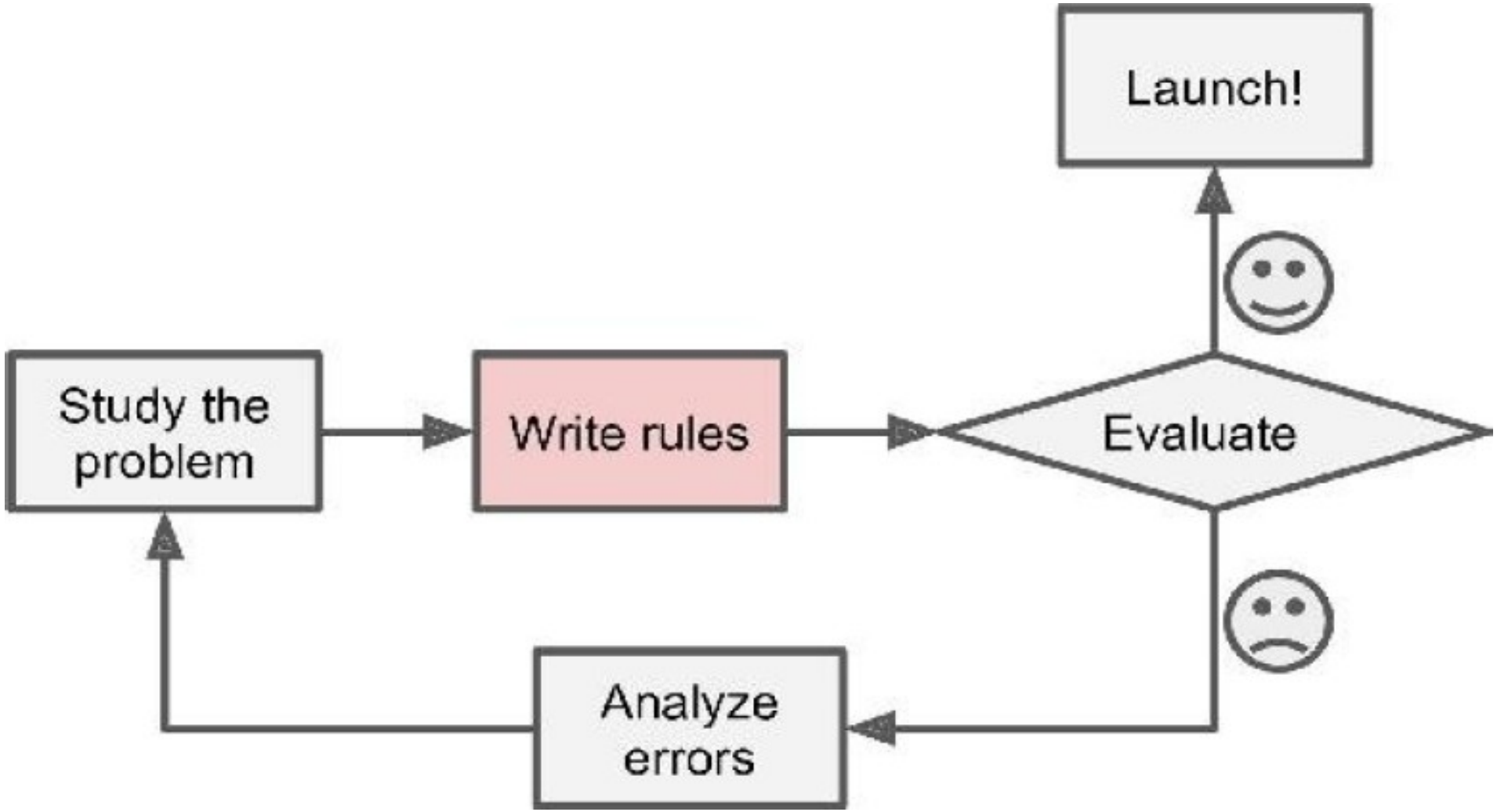


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Data-driven software development

- Two perspectives:
  - Developing data-driven software products
    - E.g. Many Artificial Intelligence (AI) applications are data-driven.
    - Also referred to as **AI Engineering**
  - Leveraging **software development data** to generate insights and build tool support for business analysts, software developers, project managers, etc.:
    - E.g. Help business analysts identify requirements from app reviews
    - E.g. Help project managers predict delays and risks
    - E.g. Help agile team estimate efforts (story points)
    - E.g. Help software developers locate security vulnerabilities and bugs
    - E.g. Automatically generate code comments, commit messages, test cases, etc.
    - Also referred to as **Software Analytics**
    - Many large organizations (e.g. Microsoft, Google and Facebook) deployed Software Analytics in their software development process.

The traditional software development approach



Source: Aurlien Gron, *Hands-On Machine Learning with Scikit-Learn and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems* (1st ed.)

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The traditional approach – an example

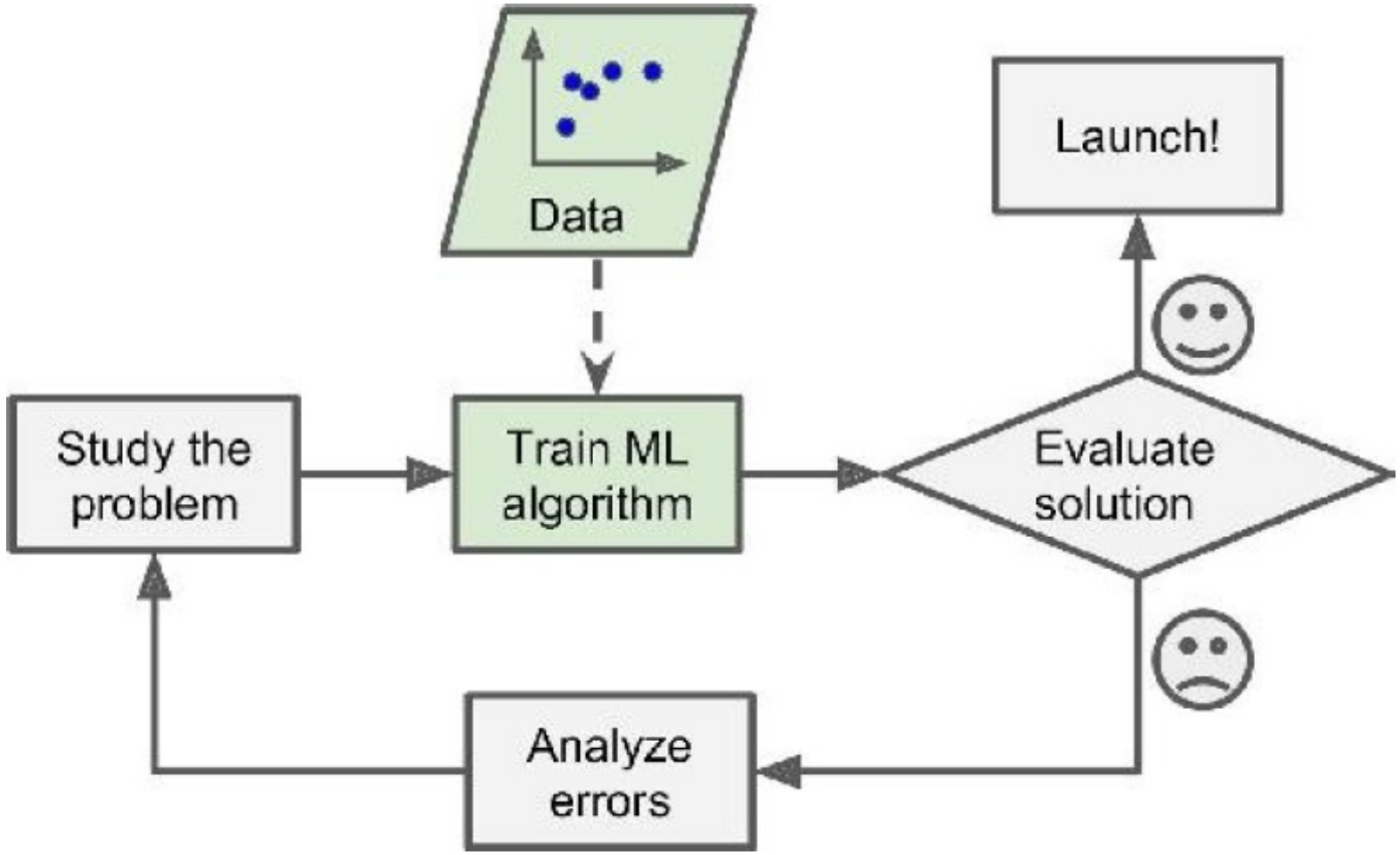
- Example: the weather problem – condition for playing a cricket game. How can we build an software app which can predict if a cricket game is going to play or not?

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	Normal	False	Yes
...	...	...	...	...

```
If outlook = sunny and humidity = high then play = no
If outlook = rainy and windy = true then play = no
If outlook = overcast then play = yes
If humidity = normal then play = yes
If none of the above then play = yes
```

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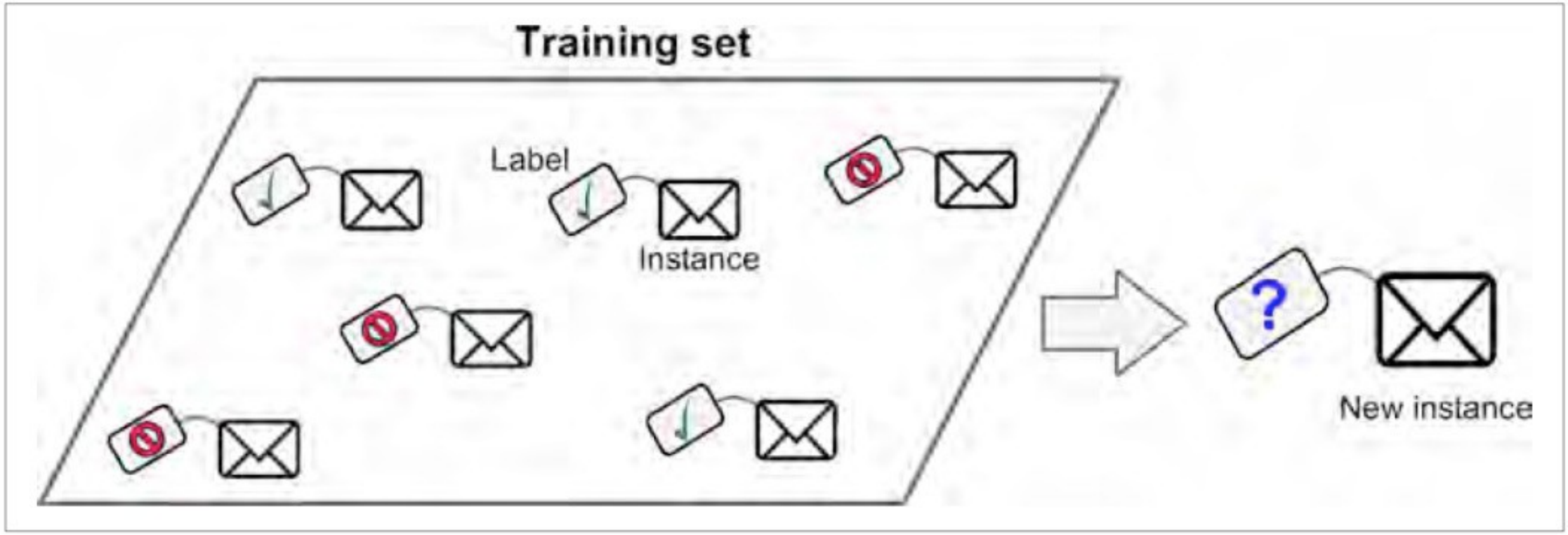
The data-driven approach



Source: Aurlien Gron, *Hands-On Machine Learning with Scikit-Learn and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems* (1st ed.)

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Data-driven software systems



Supervised learning (Machine Learning)

Source: Aurlien Gron, *Hands-On Machine Learning with Scikit-Learn and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems* (1st ed.)

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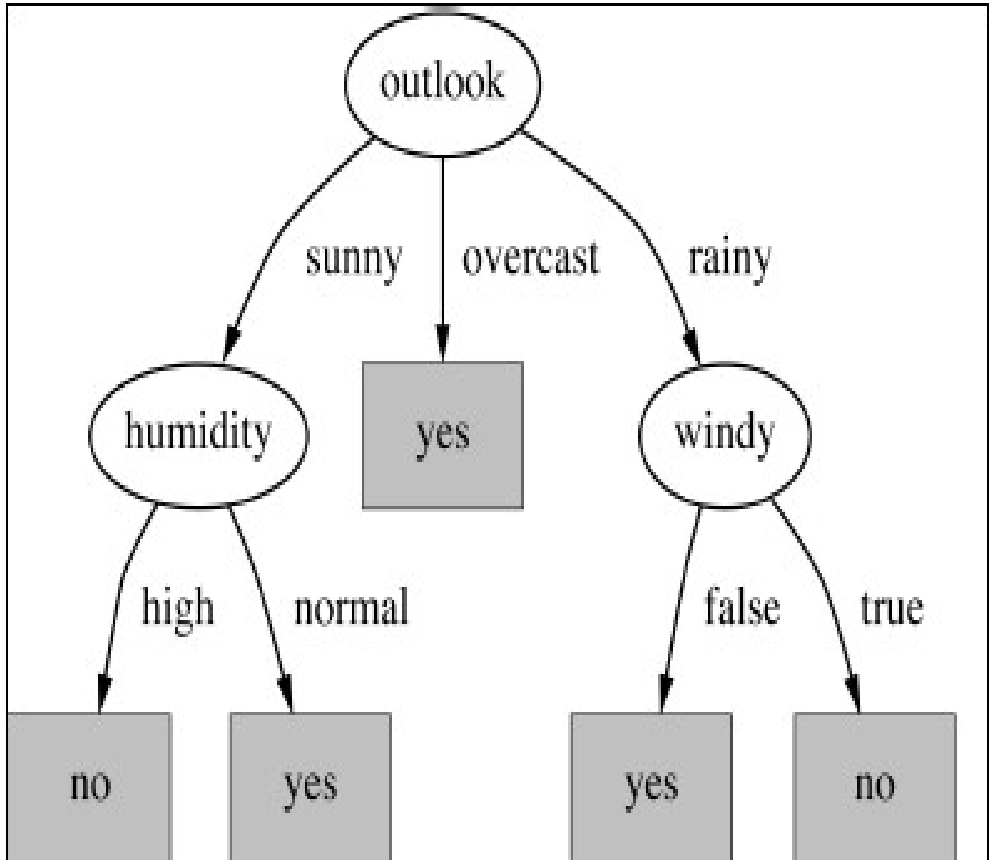
Data-driven software development
Develop data-driven learning models

- How do we (automatically) build a model (or an app) for predicting if a game is played?
  - This model is a **function** which can be inferred (learned) from labelled **training data** (aka data driven)
    - ***f(outlook, temperature, humidity, windy)*** returns true or false.
    - **Features/attributes**: input variable, e.g. outlook, tempera
    - **Target/dependent variable**, e.g. play = yes or no
    - **Training set** consists of training examples
  - Classification = prediction.

Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

The AI/ML approach – an example (cont.)

- Some basic learning models (learners)
  - **Decision Trees**



Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Source: **Data Mining: Practical Machine Learning Tools and Techniques,** *3rd Edition by Ian H. Witten, Eibe Frank, Mark A. Hall*

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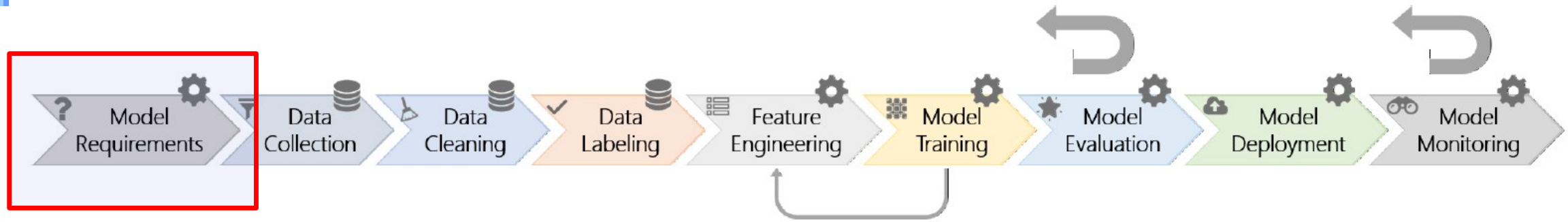
The AI/ML approach – an example (cont.)

- Some advanced learning models (learners)
  - Random Forests (RF)
    - An ensemble learning method
    - A significant improvement of the decision tree approach
    - Generating many classification trees, each of which is built with random subset of variables at each node split, and aggregates into the individual results using voting
  - Neural Networks

and many other ML models.

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Data-driven software development lifecycle

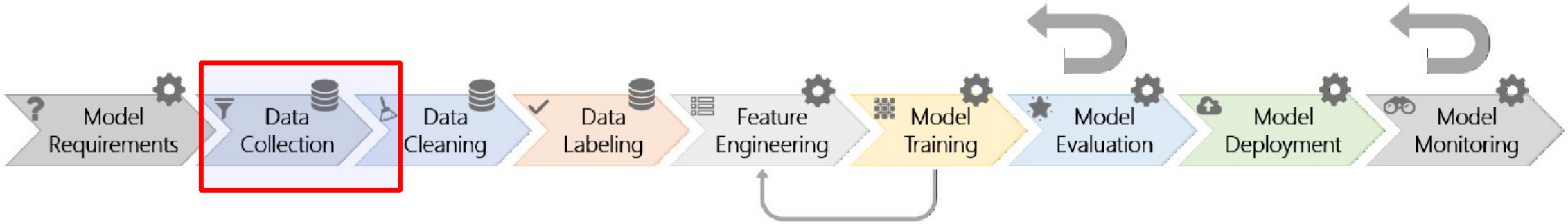


- Model requirements:
  - Identify which components of the existing (or new) product are feasible to implement with machine learning/ML, data-driven technology.
  - Elicit requirements for these data-driven/ML components
  - Determine what types of models (e.g. supervised vs. unsupervised).

Amershi et. al., Software engineering for machine learning: a case study. Proceedings of the 41st International Conference on Software Engineering: Software Engineering in Practice, 2019.

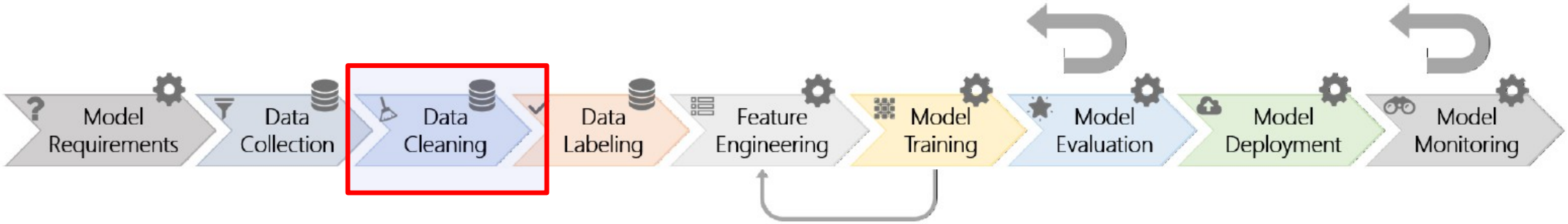
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Data-driven software development lifecycle



- Data collection:
  - Look for available datasets (e.g. internal data, public data, etc.) or build their own datasets.
  - May use a mix of datasets (e.g. pre-training on public datasets, and then (post-)training on their own dataset).

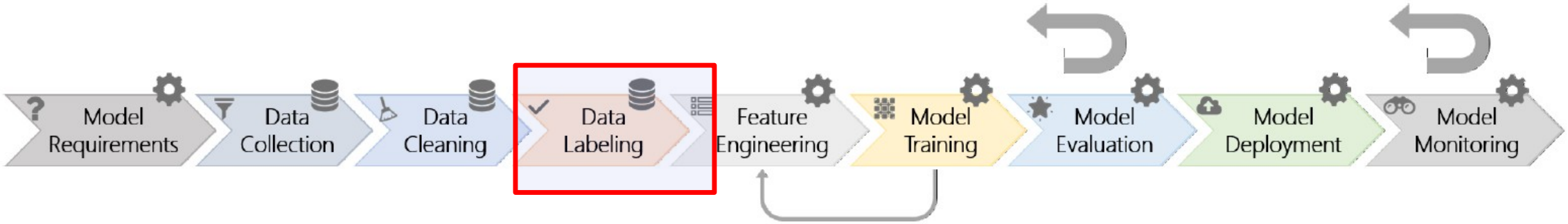
Data-driven software development lifecycle



- Data cleaning:
  - Remove inaccurate or noisy data records.
  - Filling missing data

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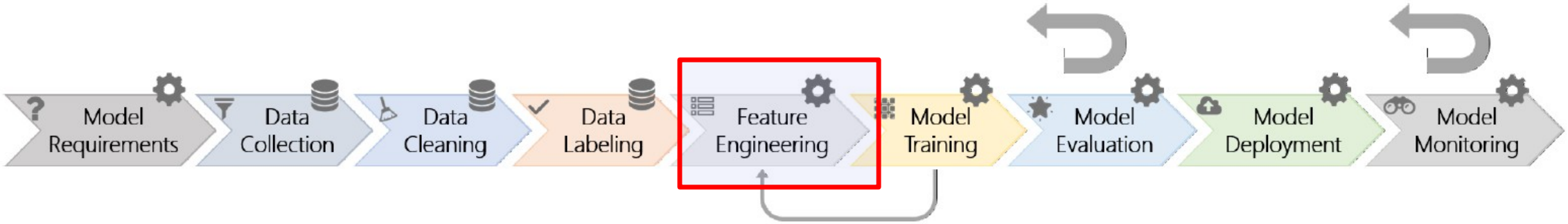
Data-driven software development lifecycle



- Data labelling:
  - Assign ground-truth labels for each data record
  - Can be done by software engineers, domain experts or crowd workers (e.g. Mechanical Turk).

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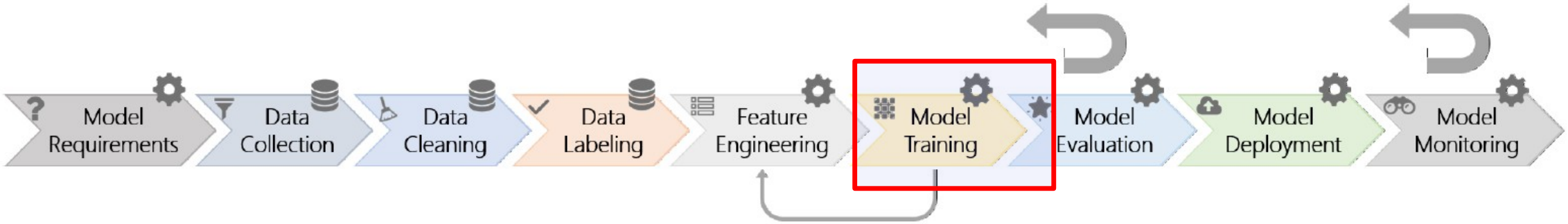
Data-driven software development lifecycle



- Feature engineering:
  - Extract features from data records – **feature extraction**
  - Select informative features (e.g. remove correlated features) – **feature selection**

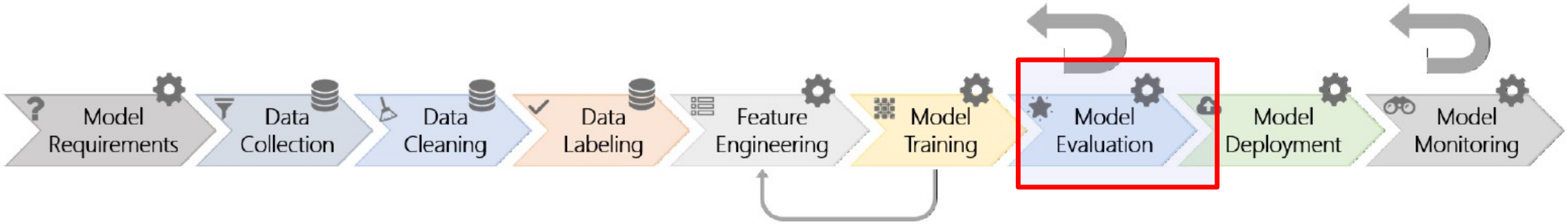
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Data-driven software development lifecycle



- Model training:
  - Split training vs test data
  - Choose a model or a set of models (learning algorithms)
  - Training the chosen models
  - Tuning hyper-parameters

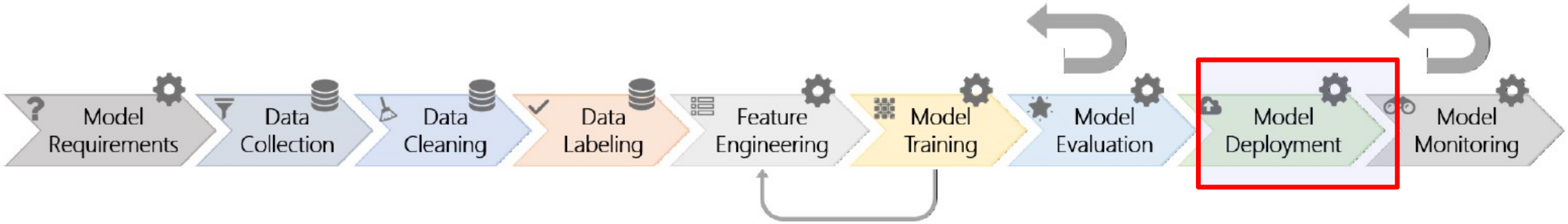
Data-driven software development lifecycle



- Model evaluation:
  - Assess the model's performance on test data (precision, recall, F-measure, AUC, MAE, etc.)

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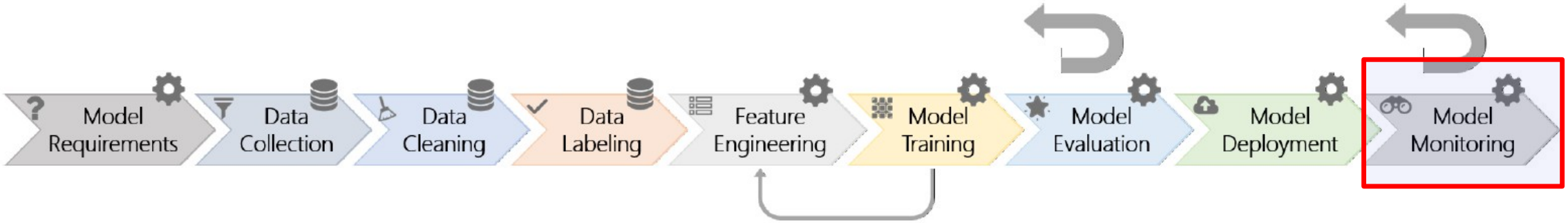
Data-driven software development lifecycle



- Model deployment:
  - Deploy on the targeted devices

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Data-driven software development lifecycle

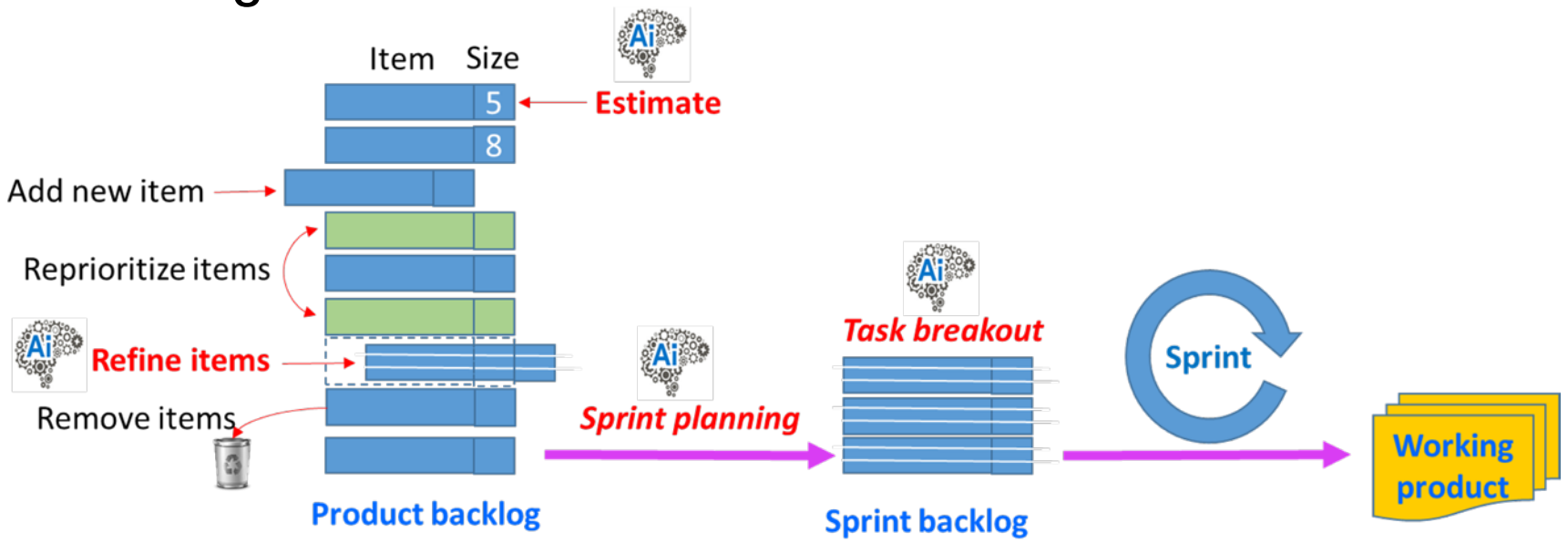


- Model monitoring:
  - Continuously monitor for performance and errors

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Software Engineering

- Requirements
- Design
- Implementation
- Verification and Validation
- Maintenance and evolution
- Software Project Management



AI for Software Engineering (AI4SE)

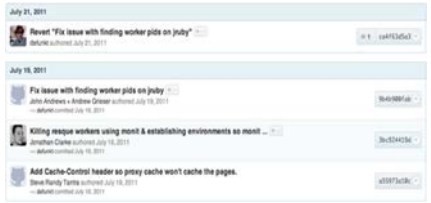




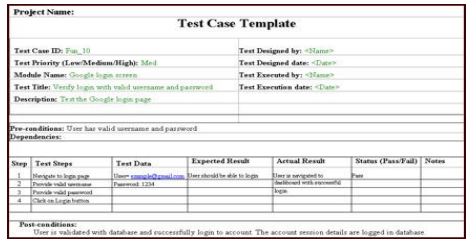
Issue, bug reports, user stories



Product backlog

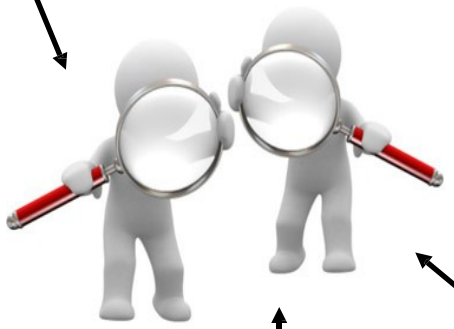


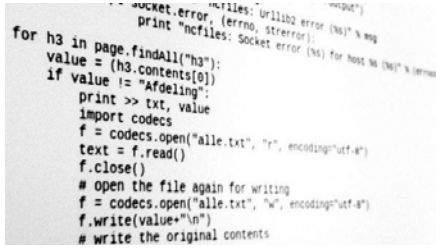
Commit messages



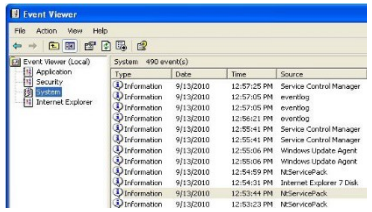
Test cases







Source code



System events

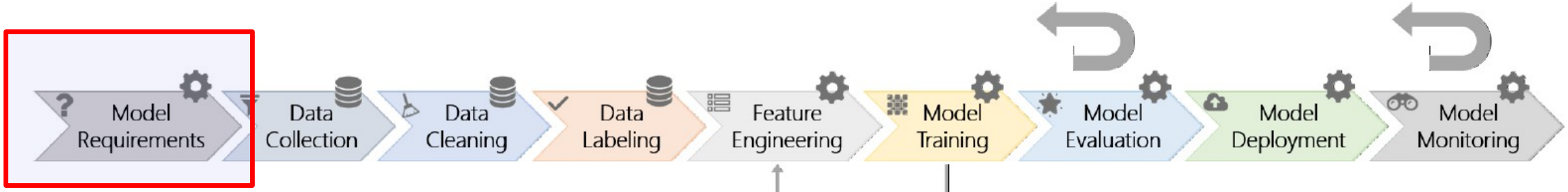


Usage logs



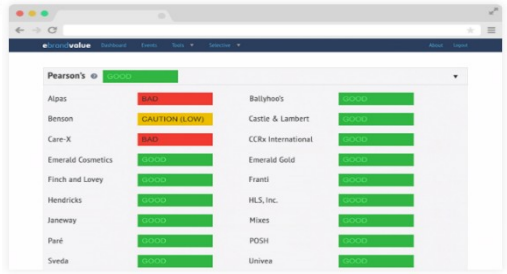
App reviews

AI4SE: delay prediction in software projects

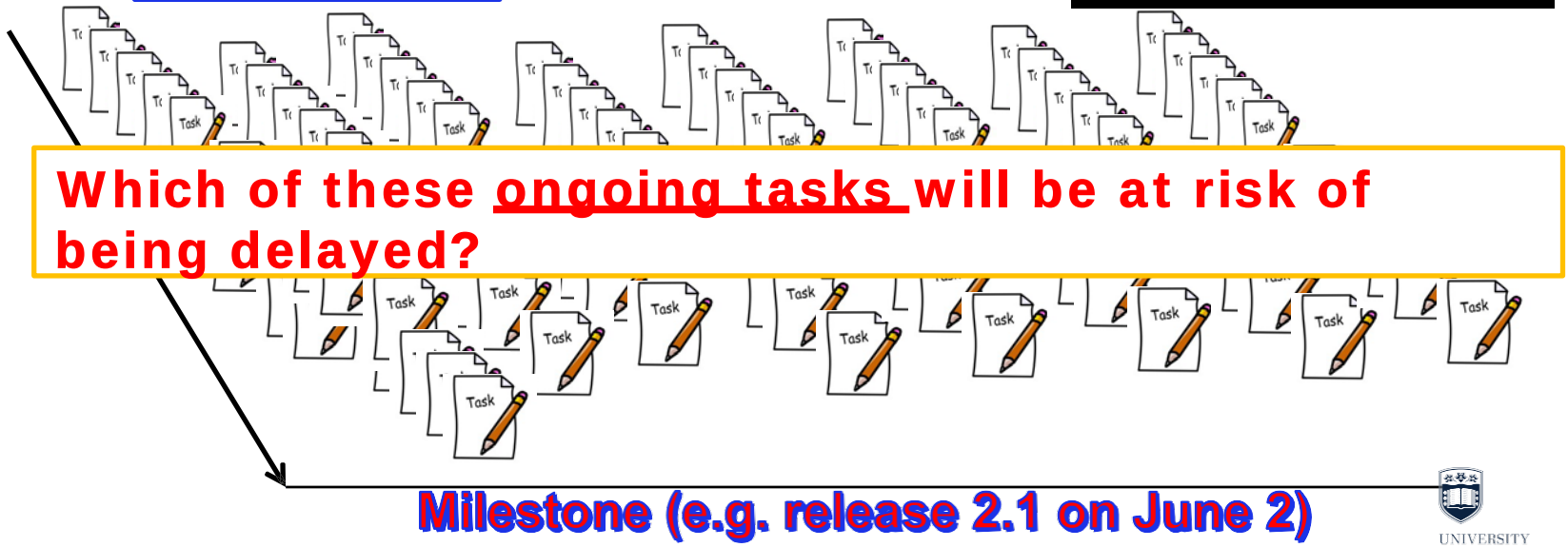


- Model requirements:



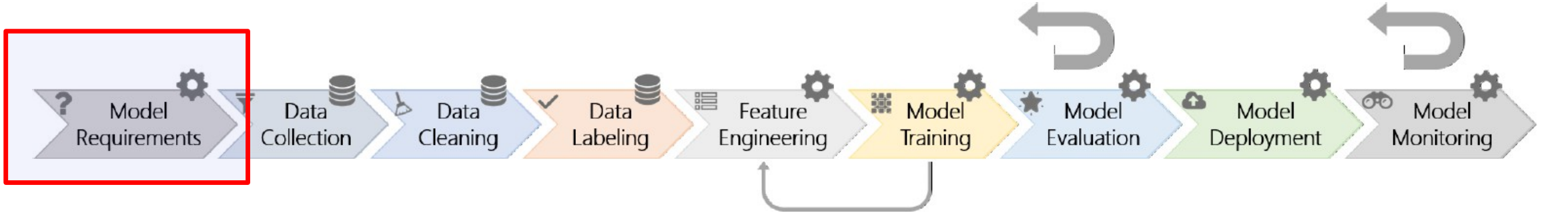




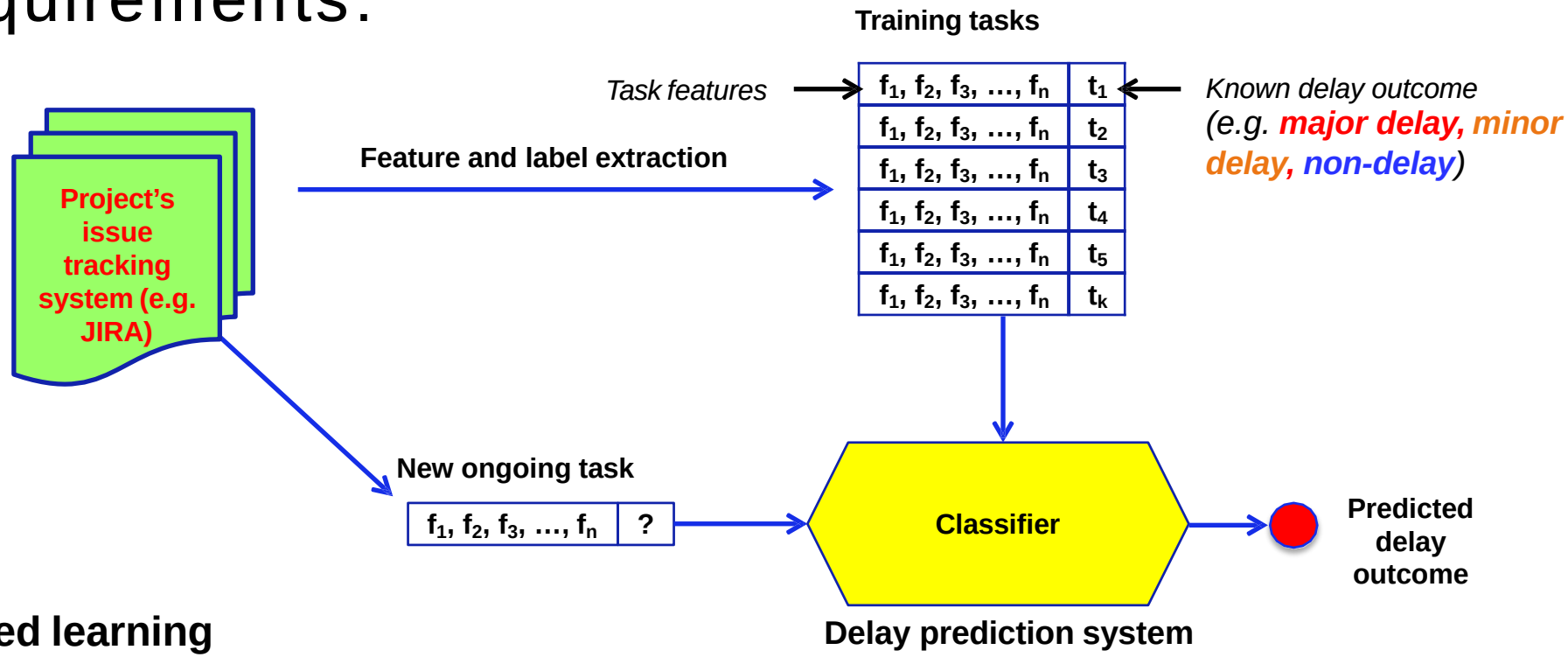


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AI Engineering lifecycle - example

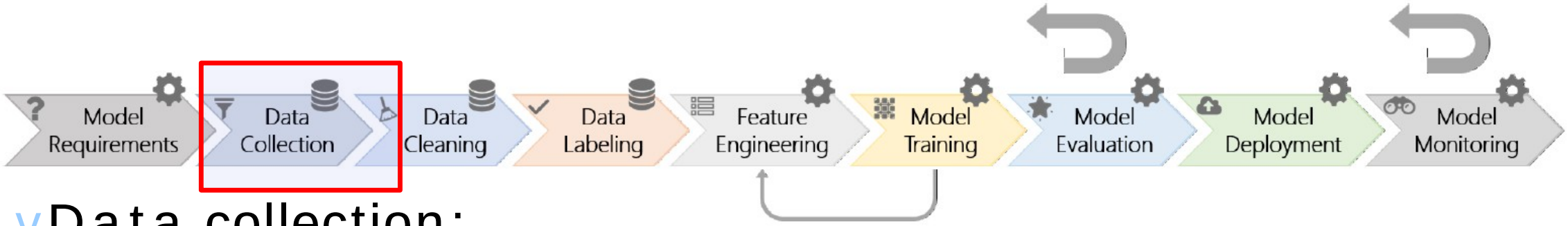


- Model requirements:



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AI Engineering lifecycle



- Data collection:
  - Analyze 40,830 past tasks (i.e. issues) in 5 large software projects: Moodle, JBoss, Apache, Duraspace, and Spring.
  - All these tasks are recorded in the JIRA issue tracking system

Application Server 3 4 5 and 6 / JBAS-13

Fix the java5 testsuite errors

Agile Board

Details

Type:	Task	Status:	CLOSED (View Workflow)
Priority:	Major	Resolution:	Done
Affects Version/s:	JBossAS-4.0.0 Final	Fix Version/s:	JBossAS-4.0.1 Final
Component/s:	None		
Labels:	None		
Environment:	> [starksm@banshee9100 testsuite] $ /usr/java/jdk1.5.0/bin/java -version java version "1.5.0" Java(TM) 2 Runtime Environment, Standard Edition (build 1.5.0-b64...		

Description

There are still some failures when running the testsuite under java5 that need to be corrected for the 4.0.1 release. This task contains the subtasks in the java5 testsuite run.

Issue Links

blocks	~~JBAS-7~~ Tasks for the 4.0.1 Release	CLOSED
is blocked by	~~JBAOP-1~~ VersionedObjectUnitTestCase fails under java5	CLOSED
	~~JBAS-14~~ CompleteUnitTestCase fails under java5	CLOSED
	~~JBAS-15~~ ContextCLTestCase fails under java5	CLOSED
	~~JBWS-52~~ MarshallDocLiteralTestCase.testEchoDateTimeArray failure under Java5	CLOSED

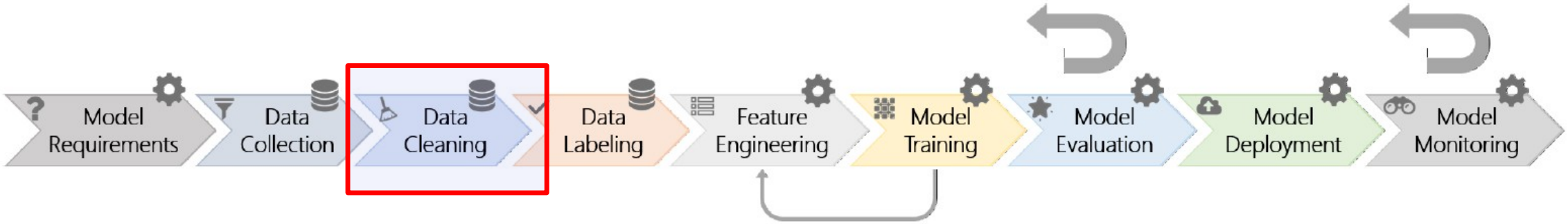
Activity

All | **Comments** | Work Log | History | Activity | Links Hierarchy

There are no comments yet on this issue.

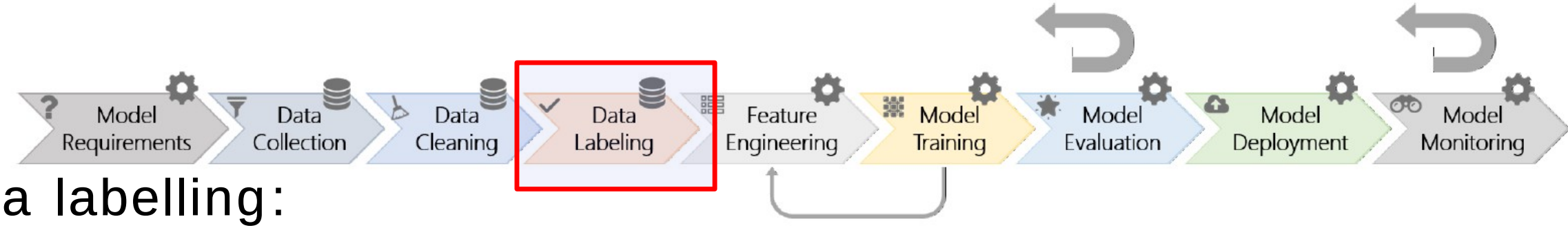
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AI Engineering lifecycle



- Data cleaning:
  - Remove outliers, e.g. incomplete tasks, issues/tasks long overdue, issues with no due date, etc.

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Data labelling:

MyFaces Core / MYFACES-2262

197 of 319

implement outcomtarget renderers

Agile Board

Export

Details

Type:	Task	Status:	CLOSED
Priority:	Minor	Resolution:	Fixed
Affects Version/s:	2.0.0-alpha	Fix Version/s:	2.0.0-alpha
Component/s:	JSR-314		
Labels:	None		

People

Assignee:	Leonardo Uribe
Reporter:	Michael Concini
Votes:	0 Vote for this issue
Watchers:	0 Start watching this issue

Dates

Due:	10/Jul/09
Created:	29/Jun/09 19:25
Updated:	07/May/10 01:58
Resolved:	10/Sep/09 16:59

Description

create link and button renderer implementations for outcome target to support bookmarkable URLs

Issue Links

is part of ~~MYFACES-2345~~ Add View Parameters feature

Agile

View on Board

Activity

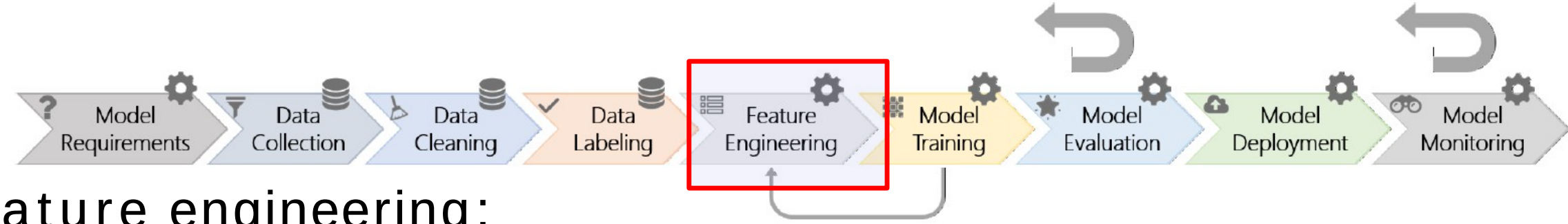
All | **Comments** | Work Log | History | Activity | Transitions

Michael Concini added a comment - 07/Jul/09 14:57

code is ready to be checked in once a new shared trunk has been created.

Leonardo Uribe added a comment - 03/Sep/09 21:05

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vFeature engineering:

MyFaces Core / MYFACES-2262

197 of 319

implement outcomtarget renderers

Agile Board

Export

Details

Type:	Task	Status:	CLOSED
Priority:	Minor	Resolution:	Fixed
Affects Version/s:	2.0.0-alpha	Fix Version/s:	2.0.0-alpha
Component/s:	JSR-314		
Labels:	None		

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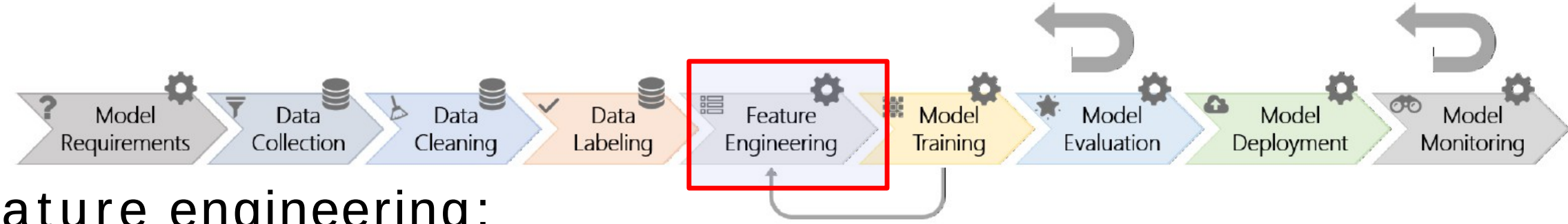
Assignee:	Leonardo Uribe
Reporter:	Michael Concini
Votes:	0 Vote for this issue
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Dates

Due:	10/Jul/09
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Agile

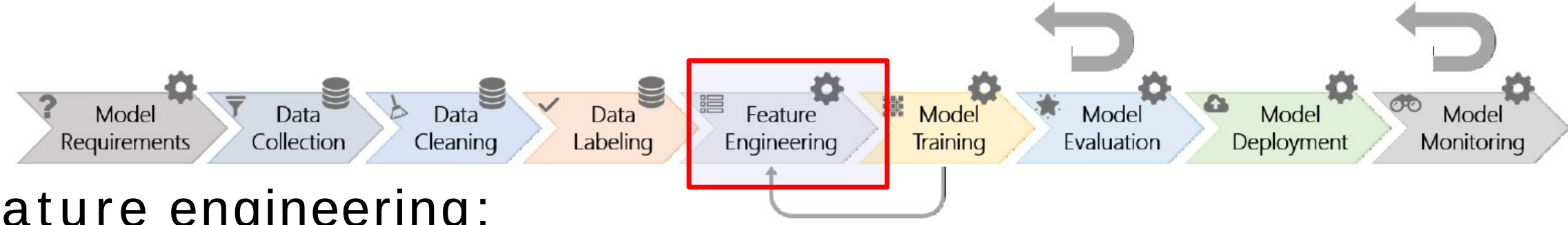
View on Board



Feature engineering:

1. Discussion time
2. Waiting time
3. Type
4. Number of times that an issue is reopened
5. Priority
6. Changing of priority
7. Number of comments
8. Number of fix versions
9. Changing of fix versions
10. Number of affect versions
11. Number of issue links
12. Number of issues that are blocked by this issue
13. Number of issues that block this issue
14. Topics of an issue's description (**NLP/LDA**)
15. Changing of description
16. Number of votes
17. Number of watches
18. Reporter reputation
19. **Developers' workload**
20. **Percentage of delayed issues that a developer involved with**
21. Task dependencies (e.g. blocking, assigned to the same person, or affecting the same components)

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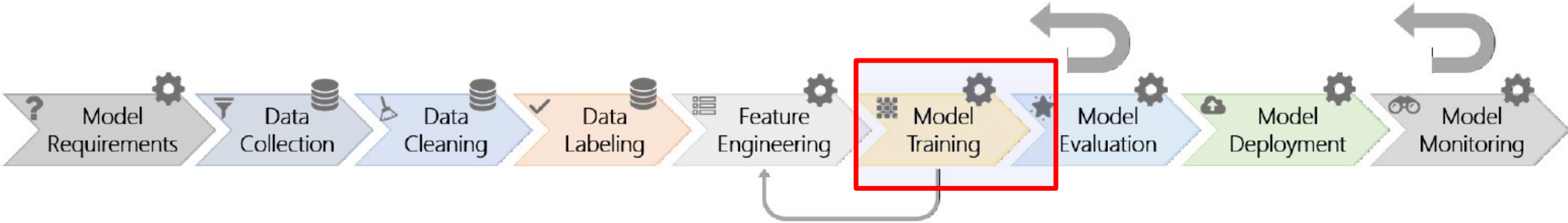


vFeature engineering:

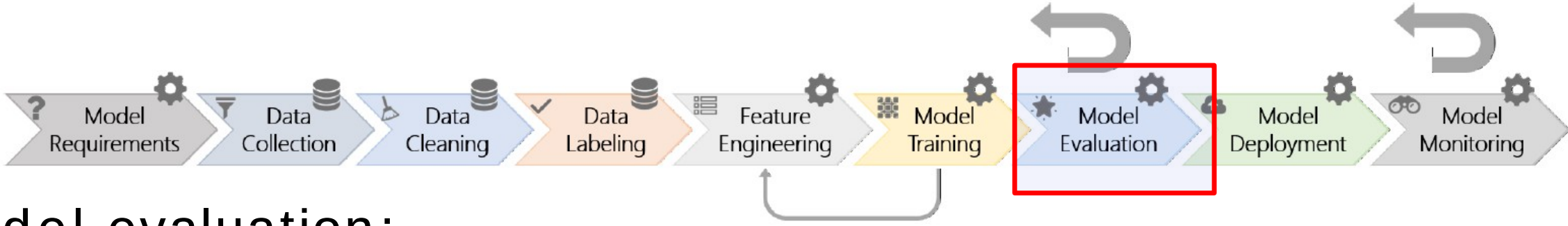
Descriptive ℓ_1-penalized logistic regression model for risk probability, trained on all tasks collected from the five projects

Feature	Apache	Duraspace	JBoss	Moodle	Spring	All
discussion	1.027	-1.368	-3.153	-21.353	-0.371	-7.274
waiting	0	0	0	-15.307	0	-4.495
type:Bug	0	-3.254	-0.959	-2.933	-4.763	-0.859
type:Code Task	N/A	-0.052	N/A	N/A	N/A	0.216
type:Documentation	0	0.359	1.617	N/A	N/A	1.06
type:Epic	0	1.026	0	0	0	0.925
type:Improvement	0.044	-0.745	N/A	-2.833	-4.704	0.271
type:New Feature	0.106	0.097	0	-3.011	-4.495	0.64
type:Story	0	-0.81	0	0	0	-0.342
type:Sub-task	-1.183	0.267	0.94	-2.796	0.001	0.867
type:Task	-0.37	-0.477	0.724	-2.604	-4.534	0.548
type:Technical task	0	0.35	0	0	0	0.567
repetition	-0.391	4.01	3.11	-0.427	6.358	2.878
perofdelay	3.183	4.202	3.088	2.569	22.091	3.361
workload	0.179	-0.911	1.367	-1.104	0.241	1.621

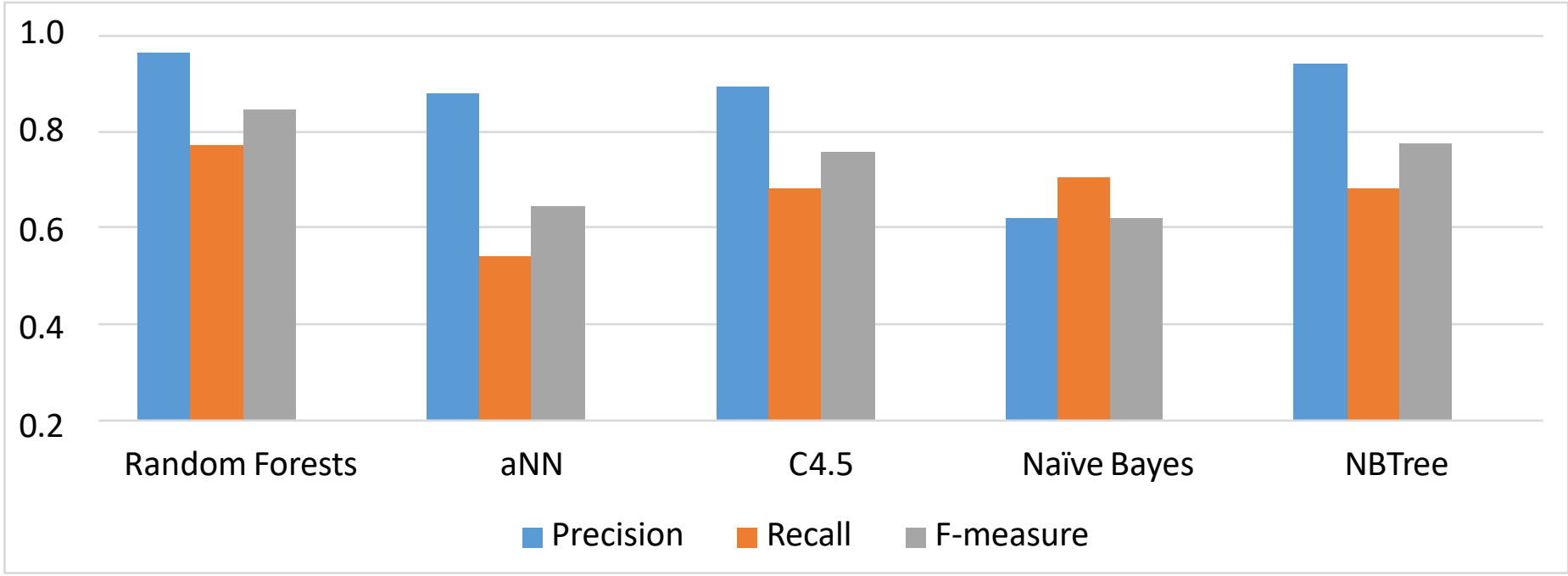
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- Model training:
  - Data (e.g. 40,830 past tasks) split into training set and test set
  - Use a number of classifiers: Random Forests, Neural Networks, Decision Tree (C4.5), Naïve Bayes and NBTree.
  - Training set is used to train these classifiers.



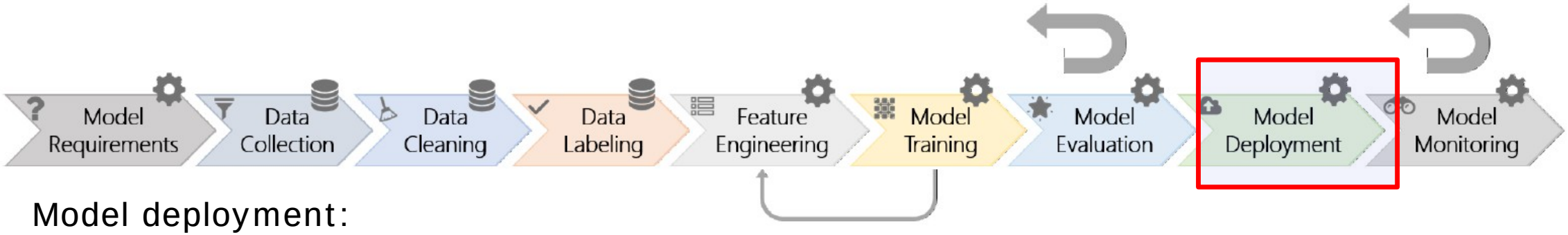
vModel evaluation:



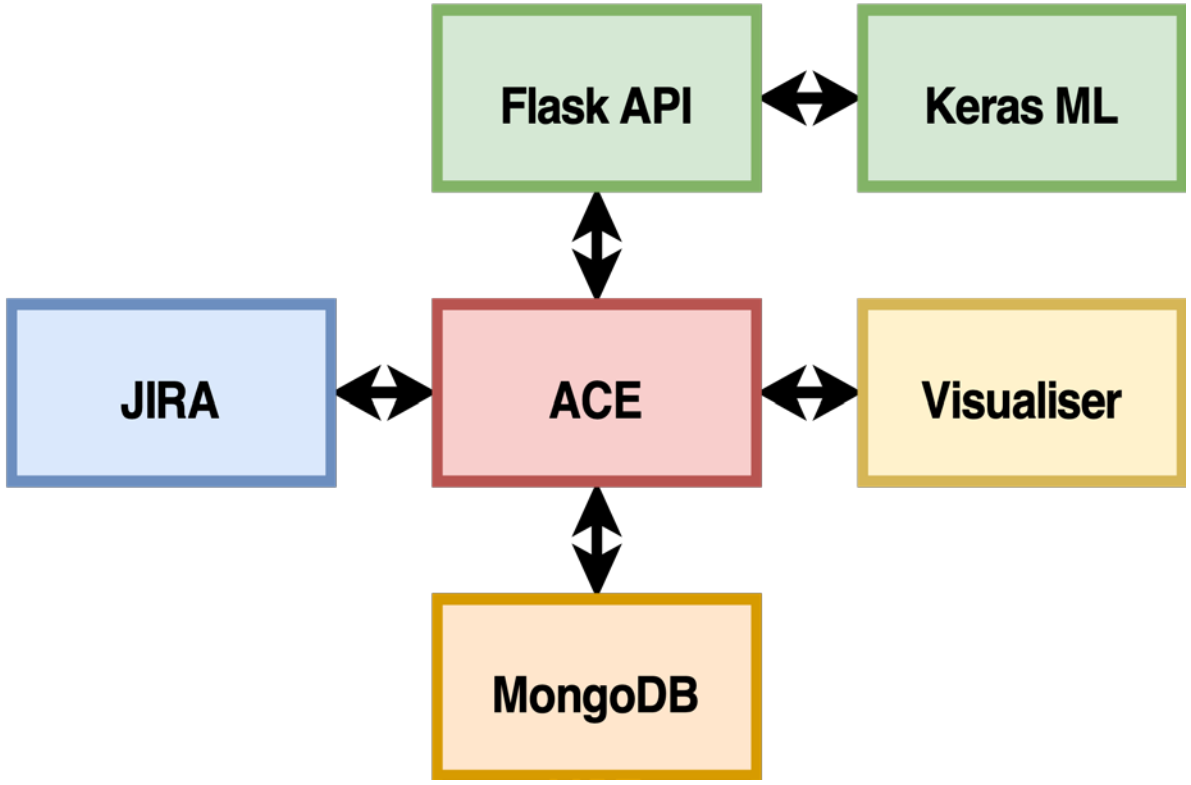
- Morakot Choetkiertikul**,** Hoa Khanh Dam, Truyen Tran and Aditya Ghose, *Characterization and prediction of issue-related risks in software projects,* Proceedings of **12th International Conference on Mining Software Repositories** (**MSR 2015**), co-located with ICSE 2015, pages 280 - 291, IEEE (ACM SIGSOFT Distinguished Paper Award)

- Morakot Choetkiertikul, Hoa Khanh Dam, Truyen Tran and Aditya Ghose, *Predicting the delay of issues with due dates in software projects,* **Empirical Software Engineering** journal, Volume 22, Issue 3, pages 1223-1263, Springer.

AI Engineering lifecycle

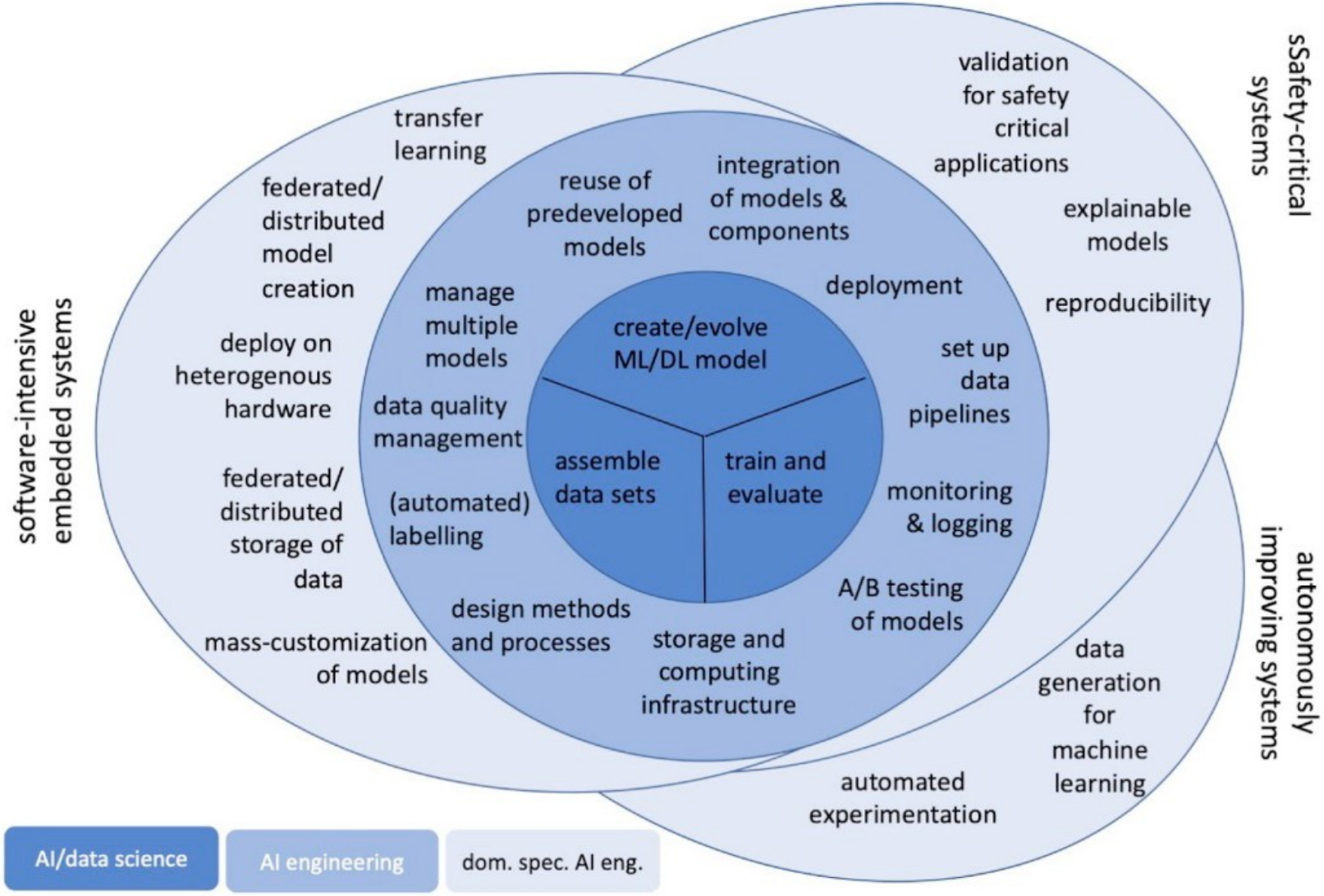


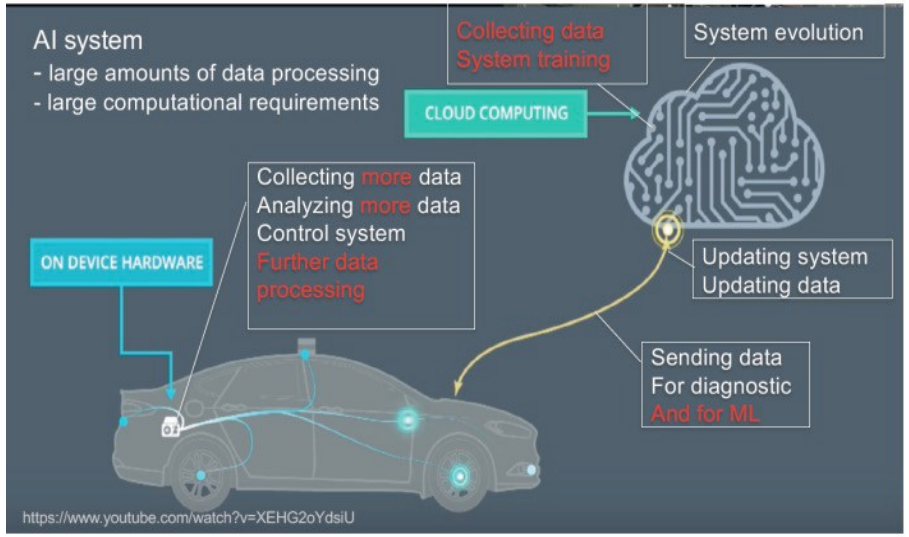
- Model deployment:
  - AI-powered plugin for JIRA issue tracking system
    - Recommending story points, labels, priority, type and components for each issue
    - Visualization of issue dependency



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Conceptualization of AI Engineering





Source: Ivica Crnkovic, *Developing AI Systems – New challenges for Software Engineering*, ICSCC, 2019

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